

Car Insurance Claim Analysis

Business Context &
Objective

Who Are Our
Policyholders ?

When and Why Do
Claim Happen ?

Which Vehicles Drive
the Most Claims ?

Who are Our Highest
Risk Customers ?

BUSINESS CONTEXT:

Accidents occur worldwide due to various factors causing significant insurance claims. Accurately estimating claims is crucial for sustainability and profitability of insurance businesses.

OBJECTIVE:

As Lead Data Analyst, this analysis explores 7,647 car insurance claim records to uncover patterns in claim frequency, claim amounts and risk profiles — enabling data-driven premium pricing and policy optimization.

DATASET:

7,647 policyholders | 21 variables |

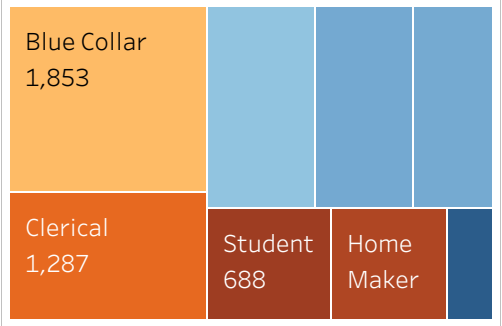
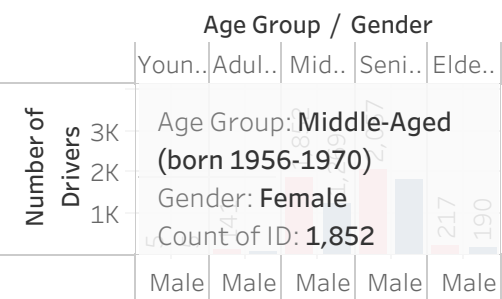
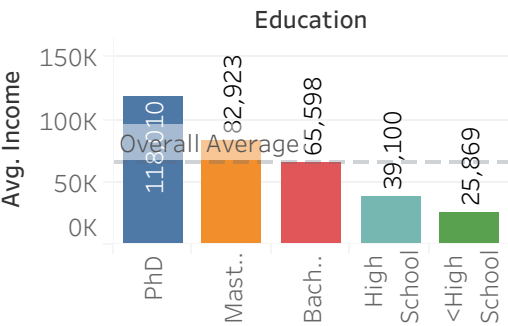
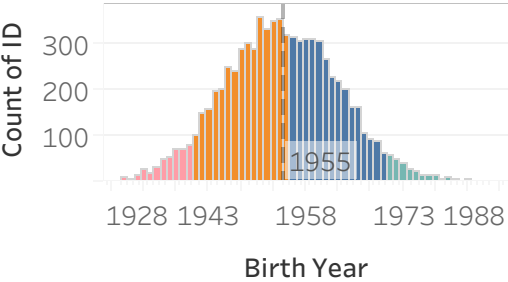
Key metrics: Claim Amount, Claim Frequency,
Bluebook Value, Education, Occupation, Car Type

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Business Context & Objective	Who Are Our Policyholders ?	When and Why Do Claim Happen ?	Which Vehicles Drive the Most Claims ?	Who are Our Highest Risk Customers ?
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Who Are Our Policyholders ?

Analysis of driver demographics, age distribution, income patterns and occupation breakdown



Gender

- Female
- Male

Age Group

- Young (born 1981+)
- Adult (born 1971-1980)
- Middle-Aged (born 1956-1970)
- Senior (born 1941-1955)
- Elderly (born 1926-1940)

Avg. Income

5,941 126,857

Education

- PhD
- Mast.
- Bach.
- High School
- <High School

Select Educ.. All

Claim Amount Filter

☒ True

Min Claim .. 0

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When and Why Do Claims Happen?

Analysis of claim amounts, frequencies and distributions by vehicle use and ge..

Gender

Female

Male

Frequency Risk Colour

High Risk

Medium Risk

Education Filter

True

Action (Education)

☒ <High School

☒ Bachelors

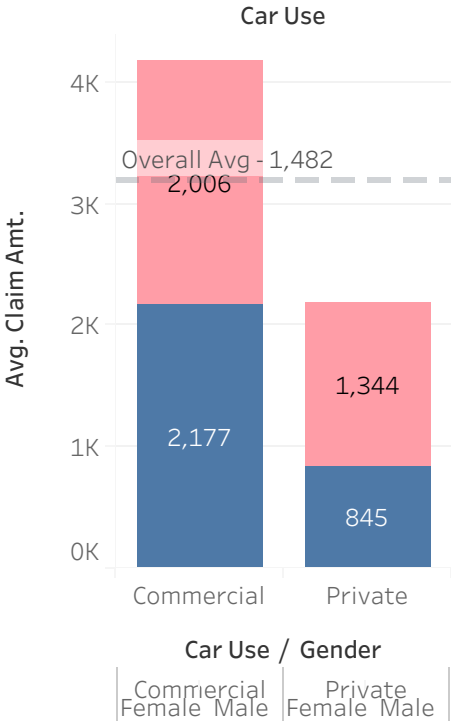
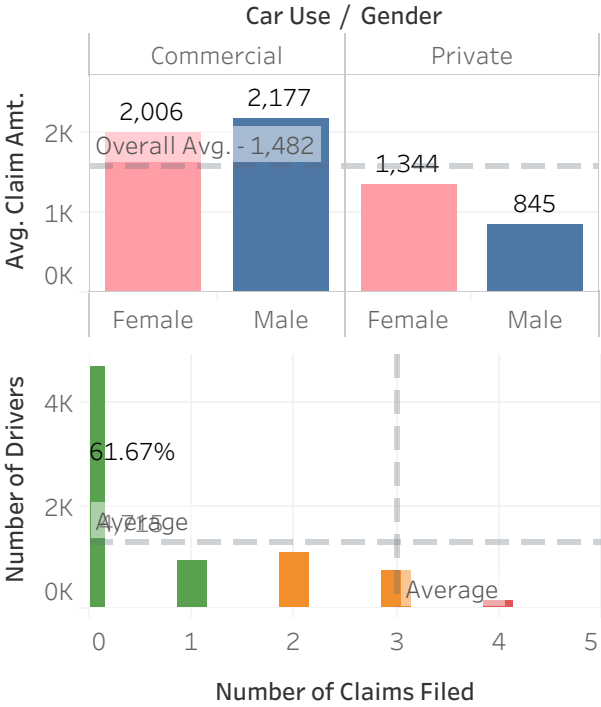
☐ High School

Claim Amount Filter

☒ True

Min Claim Amount

0

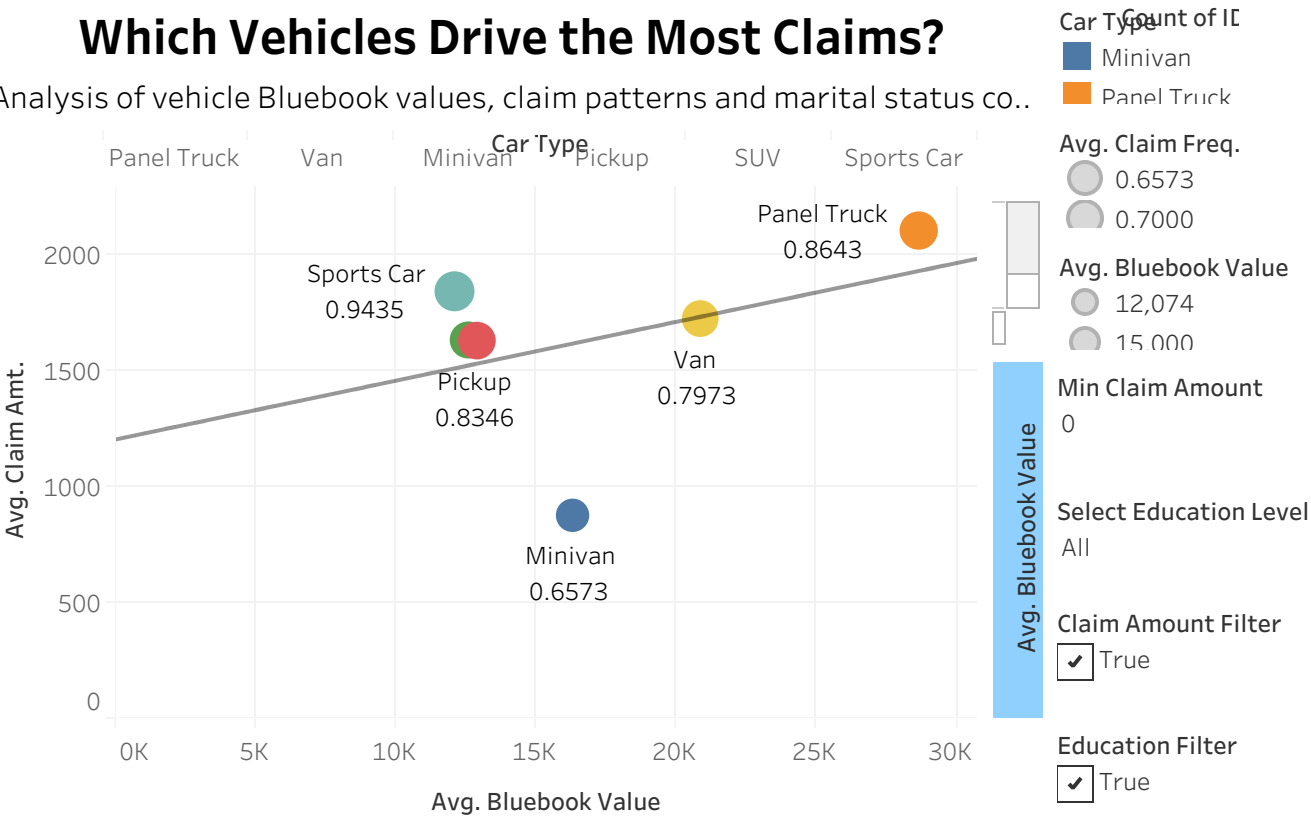


Car Insurance Claim Analysis

Who Are Our Policyholders ?	When and Why Do Claim Happen ?	Which Vehicles Drive the Most Claims ?	Who are Our Highest Risk Customers ?	Where are Pricing Opportunities ?
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Which Vehicles Drive the Most Claims?

Analysis of vehicle Bluebook values, claim patterns and marital status co..



Car Insurance Claim Analysis

When and Why Do Claim Happen ?	Which Vehicles Drive the Most Claims ?	Who are Our Highest Risk Customers ?	Where are Pricing Opportunities ?	Executive Summary.
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Who Are Our Highest Risk Customers ?

Analysis of driver risk profiles, occupation claim patterns and geographic risk distribution

Risk Score

High Risk

Measure Names

Claim Amt.

Avg. Claim Amt.

Avg. Claim Freq.

0.000

Education Filter

True

Select Education Level

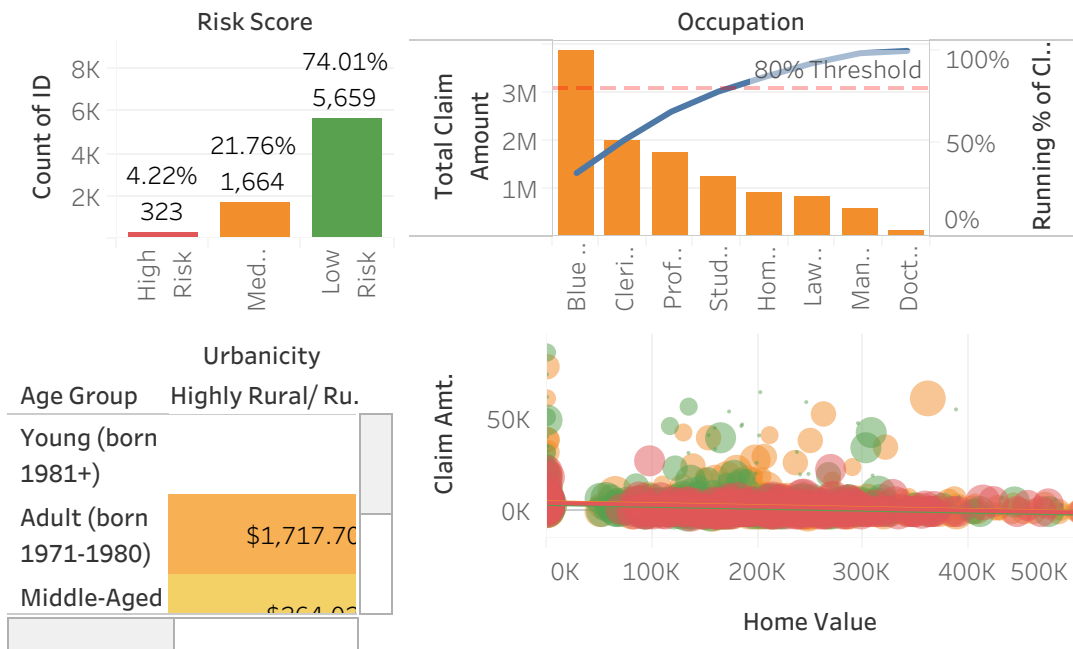
All

Claim Amount Filter

True

Min Claim Amount

0



Car Insurance Claim Analysis

When and Why Do Claim Happen ?

Which Vehicles Drive the Most Claims ?

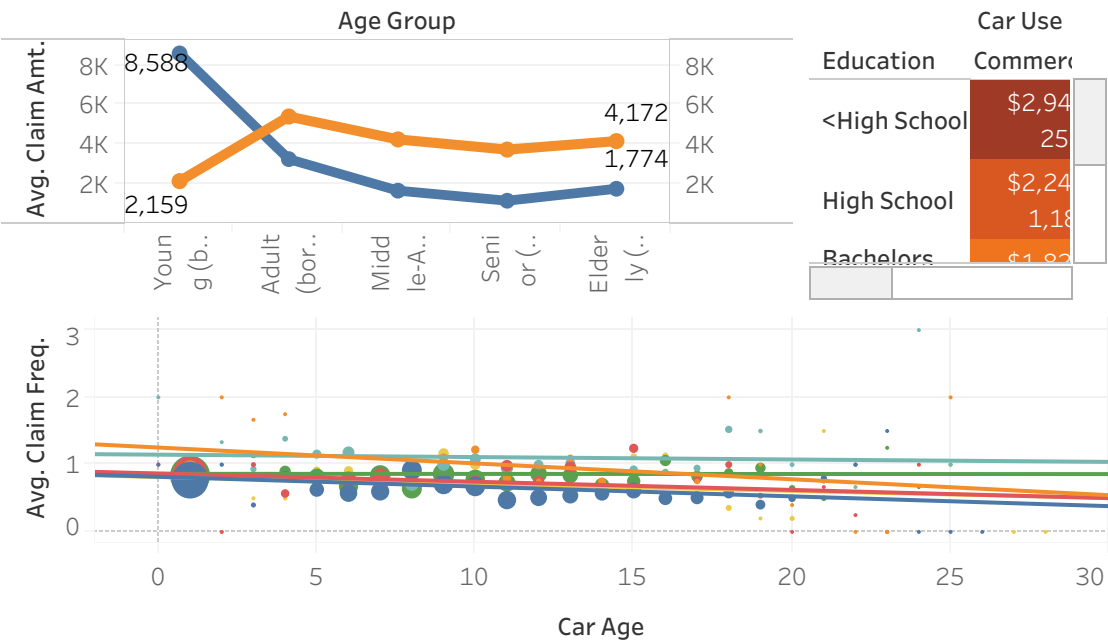
Who are Our Highest Risk Customers ?

Where are Pricing Opportunities ?

Executive Summary.

Where Are the Pricing Opportunities ?

Analysis of claim trends across age groups, education levels and vehicle usage patterns



Measure N.. Avg. Clai..

Car Type Minivan

Avg. Claim ..

Count of ID 1

Education .. ☒ True

Select Education Level

☒ All

☐ <High School

☐ Bachelors

☐ High School

☐ Masters

☐ PhD

Claim Amo.. ☒ True

Min Claim .. 0

Car Insurance Claim Analysis

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Car Insurance Claims Analysis - Executive Summary

KEY FINDINGS FROM ANALYSIS:

Dashboard 1 — Driver Demographics:

- Middle-Aged Female drivers (born 1956-1970) form the largest policyholder segment with 1,852 drivers.
- PhD holders earn \$118,010 avg income — 4.6× more than below High School drivers (\$25,869). Clear income ladder exists across all education levels.

Dashboard 2 — Claims Deep Dive:

- Commercial Male drivers file highest avg claims (\$2,177) — 2.6× higher than Private Male drivers (\$845).
- 61.67% of all drivers (4,715) have zero claims. Only 4.22% (323 drivers) are High Risk yet drive disproportionate claim exposure.

Dashboard 3 — Vehicle Profile:

- Panel Trucks have highest Bluebook value (\$28,630) exceeding all other vehicle types including Sports Cars.
- SUVs are most popular among both Married (1,354) and Single (910) drivers. Positive correlation confirmed between vehicle value and claim amount.

Dashboard 4 — Risk Profiling:

STRATEGIC RECOMMENDATIONS:

1. PREMIUM PRICING: Implement tiered premium pricing using Risk Score (High/Medium/Low) based on CLM_FREQ and CAR_AGE to maximize profitability.
2. COMMERCIAL VEHICLE POLICY: Increase deductibles for Commercial vehicle policyholders especially in Highly Urban areas where claim amounts are highest.
3. CUSTOMER RETENTION: Offer loyalty discounts to Low Risk drivers (74% of portfolio) to reduce churn in this highly profitable segment.
4. HIGH VALUE VEHICLES: Develop specialized claim management protocols for Panel Trucks and high Bluebook vehicles to cap loss exposure.